The Higgs boson in the four-lepton channel with ATLAS Open Data

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This note analyses the decay $H \to ZZ^* \to 4\ell$ with the ATLAS Open Data for education (13 TeV, 2015–2016, 36.6 fb^{-1}) [1]. Starting from the official ATLAS notebook [2], it measures the Higgs boson mass, computes the significance of the signal with a profile likelihood, and tests whether a simple classifier improves the result.

1 Data and selection

Events with exactly four leptons are compared with simulation of the Higgs signal ($m_H = 125$ GeV), the ZZ^* background and smaller backgrounds. Each simulated event is weighted by luminosity, cross section and detector scale factors. The selection first reproduces the official notebook exactly, then pairs the leptons into two Z candidates (same flavour, opposite charge), with $50 < m_{12} < 106$ GeV and $m_{34} > 12$ GeV. Figure 1 shows the four-lepton mass: the Higgs boson appears as a narrow peak near 125 GeV above the ZZ^* background.

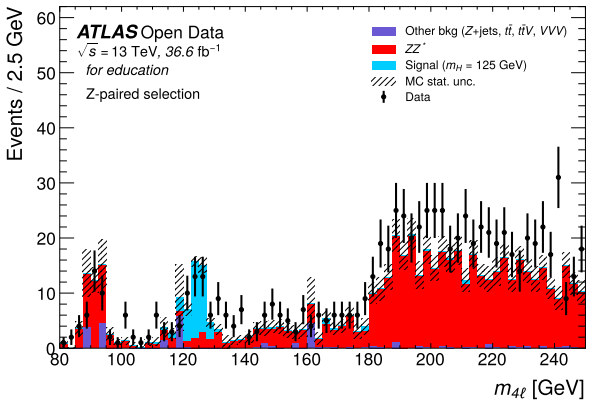


Figure 1: Four-lepton invariant mass: data and stacked simulation.

2 Higgs boson mass

The 84 data events with $110 < m_{4\ell} < 140$ GeV are fitted with an unbinned maximum-likelihood fit: a Crystal Ball peak for the signal, with its shape taken from simulation, on top of an exponential background. The method was first tested on 2000 simulated pseudo-experiments, which showed an unbiased result with correctly estimated uncertainties. The fit to data gives

$$m_H = 125.06^{+0.93}_{-0.92}\,\text{GeV} \quad \text{(statistical uncertainty)}, \tag{1}$$

in agreement with the ATLAS measurement in this channel, 124.79 ± 0.37 GeV [3], and with the world average, 125.13 ± 0.11 GeV [4].

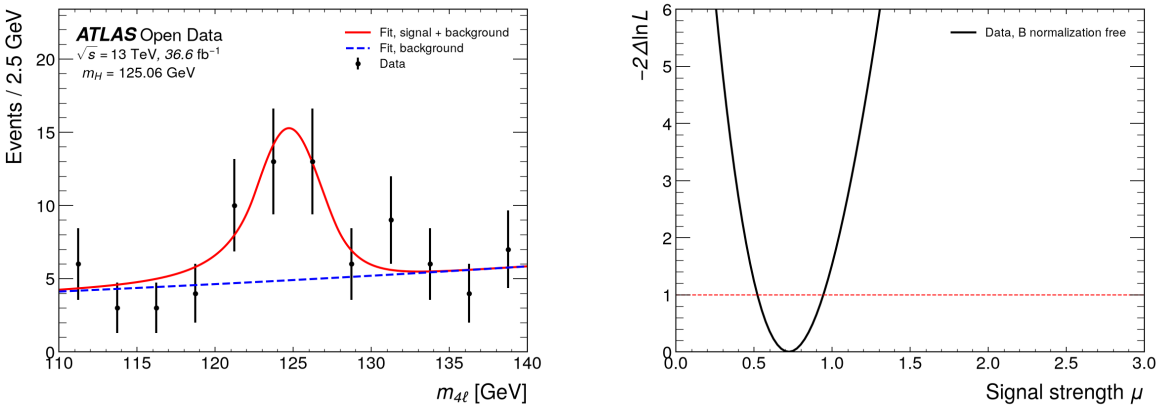


Figure 2: Left: mass fit to data. Right: likelihood scan of the signal strength μ.

3 Significance

The four-lepton mass spectrum between 110 and 160 GeV is fitted with pyhf [5], with the signal strength μ as the parameter of interest and a free background normalization. The observed significance is 3.96σ, for 6.64σ expected, and the fitted signal strength is $\hat{\mu} = 0.72^{+0.22}_{-0.20}$ (Figure 2, right). Table 1 shows how simple estimates compare with the full likelihood fit.

Method	Expected significance
$S/\sqrt{B}$ in 122.5–130 GeV	11.2
Counting experiment, Asimov formula [6]	7.8
Likelihood fit of the mass spectrum	6.6

Table 1: Expected significance with three methods. The observed value from the full fit is 3.96σ.

4 Classifier

A logistic-regression classifier was trained on ten lepton kinematic variables, excluding the four-lepton mass, to separate signal from background. It was trained on half of the simulation and tested on the other half, reaching an area under the ROC curve of 0.70. Splitting the events into two categories by classifier score did not improve the expected significance (5.97 against 6.06 without categories).

5 Conclusion

With 36.6 fb^{-1} of open ATLAS data, the four-lepton channel gives $m_H = 125.06^{+0.93}_{-0.92}$ GeV and an observed signal significance of 3.96σ. The comparison of methods shows that a full likelihood fit gives a lower, more realistic significance than simple formulas.

References

[1] ATLAS Collaboration, ROOT ntuple format 2015–2016 proton-proton Open Data for Education and Outreach beta release from the ATLAS experiment, CERN Open Data Portal (2025), doi:10.7483/OPENDATA.ATLAS.B5M9.44TN.

[2] ATLAS Open Data, `HZZAnalysis.ipynb`, `https://github.com/atlas-outreach-data-tools/notebooks-collection-opendata`.

[3] ATLAS Collaboration, Measurement of the Higgs boson mass in the $H \to ZZ^* \to 4\ell$ and $H \to \gamma\gamma$ channels with $\sqrt{s} = 13\,\text{TeV}$ pp collisions using the ATLAS detector, Phys. Lett. B 784 (2018) 345, arXiv:1806.00242.

[4] Particle Data Group, Higgs boson mass, PDG Live (2026 edition), `https://pdglive.lbl.gov/view/S126M`.

[5] L. Heinrich, M. Feickert, G. Stark, K. Cranmer, pyhf: pure-Python implementation of HistFactory statistical models, J. Open Source Softw. 6 (2021) 2823.

[6] G. Cowan, K. Cranmer, E. Gross, O. Vitells, Asymptotic formulae for likelihood-based tests of new physics, Eur. Phys. J. C 71 (2011) 1554, arXiv:1007.1727.